Power & Actuation PCB (Board 1)

MAGTRACK magnetic wall-climbing pipeline-inspection robot · Capstone Design Project, Higher Colleges of Technology
Designed by Elham Mohammed Kedir (EasyEDA, milled in-house) · Board 2 (logic) was designed by a teammate

1. Purpose

Board 1 turns the 24 V battery into the 5 V and 3.3 V rails and switches the loads the controller must not drive directly: a hardware motor-enable relay, the couplant pump and the probe servo supply. **No motor current flows through the board**; the battery feeds the BTS7960 motor drivers directly. A 10-pin stack connector joins it to the Arduino Mega logic board (Board 2).

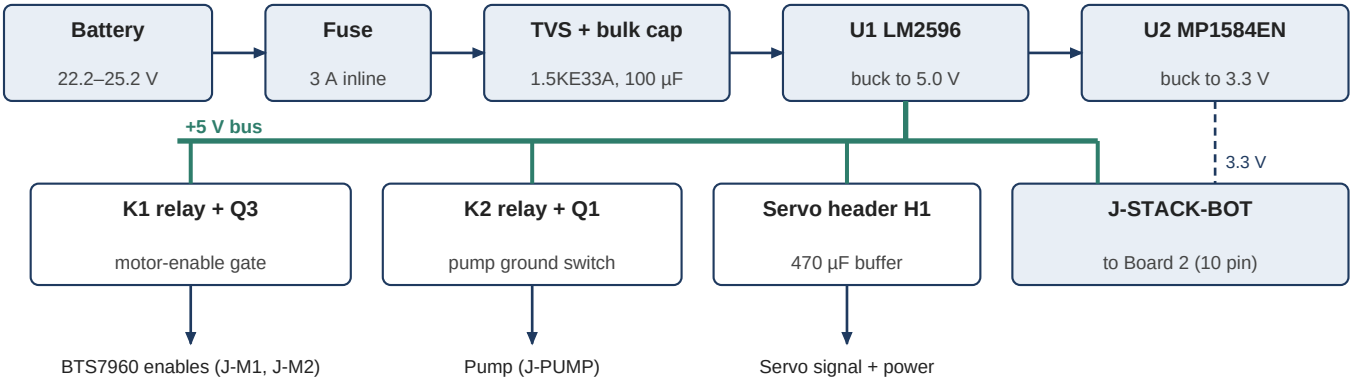


Figure 1. Power and signal flow on Board 1. Control signals (SAFETY_DRIVE, PUMP_DRIVE, SERVO, motor PWM) arrive from Board 2 through J-STACK-BOT.

2. Design features

Function	Implementation	Purpose
Input protection	3 A inline fuse on the board feed; 1.5KE33A TVS and 100 µF across the input	Limits fault current into the traces; clamps spikes from motor braking and hot-plugging
5 V rail	LM2596 buck module, set to 5.0 V	Logic, relay coils, pump and servo
3.3 V rail	MP1584EN buck module fed from the 5 V rail	Sensors on Board 2; avoids stepping 24 V straight down
Motor safety gate	K1 relay (SRD-05VDC-SL-C) driven by Q3 (2N7000); R5 1 kΩ gate, R3 10 kΩ gate pull-down, R7 10 kΩ pull-down on MOTOR_ENABLE_SAFE	Relay is off at boot, reset or power loss, so the driver enable pins read low
Pump switch	K2 relay driven by Q1 (2N7000); R6 1 kΩ, R4 10 kΩ; pump ground switched through K2	Pump runs only while PUMP_DRIVE is high
Inductive protection	D1, D2 (1N4007) across the relay coils; D3 (1N4007) across the pump terminals	Clamps coil and motor kick-back
Servo supply	Dedicated header H1 with a 470 µF capacitor beside it	Absorbs servo current surges so they do not pull down the logic rail

Fail-safe logic. The controller must hold SAFETY_DRIVE high to keep K1 closed and MOTOR_ENABLE_SAFE at 5 V. A reset, a floating pin or loss of power drops the relay and the pull-down takes the enable lines to 0 V, which stops both drive motors.

3. Interfaces

Pin	Net	Function
1	GND	Ground
2	+5V	5 V to Board 2
3	+3.3V	3.3 V to Board 2
4	SAFETY_DRIVE	Motor-enable relay (K1) control
5	PUMP_DRIVE	Pump relay (K2) control
6 / 7	M1_RPWM / M1_LPWM	Motor driver 1 PWM
8 / 9	M2_RPWM / M2_LPWM	Motor driver 2 PWM
10	SERVO	Servo PWM to header H1

Ref	Type	Function
J-PWR	2-pin screw terminal	Fused battery input
J-STEP-PWR	2-pin screw terminal	Battery tap for the stepper driver
J-PUMP	2-pin screw terminal	5 V and switched ground to pump
J-M1, J-M2	2×4 header	Enable, PWM and 5 V to each BTS7960
H1	1×3 header	Servo signal, +5 V, GND
J-STACK-BOT	1×10 female header	Link to Board 2

4. Fabrication

- Milled on the campus CNC router: 20 mil clearance, 20 mil minimum signal traces, 40–60 mil power traces.
- All through-hole parts; pads 1.8 mm or less (2.0–2.2 mm for the TVS and screw terminals) to meet the lab's 72 mil limit.

5. Bring-up checklist and status

- Check 5.0 V and 3.3 V with nothing connected.
- Confirm both relays stay open at power-up.
- Confirm the BTS7960 enable pins read about 0 V with K1 open.
- Measure the peak 5 V load (servo, pump, relays, logic) against the LM2596 module rating before a full run.
- Connect Board 2 last.

Status: layout finalized. Fabrication and test results will be added to the repository once recorded.